

Problem 8.20

A bank offers a 272-day discounted loan at a simple discount rate of 12%.

- (a) How much money would a borrower receive if she asked for a \$5000 loan?
- (b) What size loan should the borrower ask for in order to actually receive \$5000?
- (c) What is the equivalent simple interest rate that is being charged on the loan?

Solution

The simple discount rate is

$$d = 12\% = 0.12, \quad t = \frac{272}{365}.$$

The timeline is:



(a)

For a simple discount loan, the amount received is

$$X = F(1 - dt),$$

where F is the face amount of the loan. If $F = 5000$, then

$$\begin{aligned} X &= 5000 \left(1 - 0.12 \left(\frac{272}{365} \right) \right) \\ &\approx 4552.876712. \end{aligned}$$

Thus, the borrower would receive

$$\boxed{\$4,552.88}.$$

(b)

Now the borrower wants to receive \$5000. Let Y be the face amount of the loan. Then

$$5000 = Y \left(1 - 0.12 \left(\frac{272}{365} \right) \right).$$

Solving for Y ,

$$Y = \frac{5000}{1 - 0.12 \left(\frac{272}{365} \right)} \\ \approx 5491.038819.$$

Thus, the borrower should ask for

$$\boxed{\$5,491.04}.$$

(c)

For simple discount and simple interest over the same time period, the equivalent simple interest rate is

$$i = \frac{d}{1 - dt}.$$

Therefore,

$$i = \frac{0.12}{1 - 0.12 \left(\frac{272}{365} \right)} \\ \approx 0.131784932.$$

So the equivalent simple interest rate is

$$\boxed{i \approx 13.18\%}.$$